

Many organisations are rapidly adopting AI through their machine learning platforms built on TensorFlow, PyTorch and others. These platforms enhance software development, testing and the delivery teams' speed and agility.

Change is never easy and is often met with scepticism. Yet, high-performing organisations adopt and redefine how they use technology and manual interventions by automating every possible process. While AI promises to minimise developer functions, there are some worries about loss of jobs. However, by embracing AI with proper planning and establishing governance policies, organisations can enhance decision-making, efficiency and productivity, without necessarily eliminating human roles. **END** 

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Disclaimer: This article expresses the views of the author and not of the organisation he works in.

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DVC (Data Version Control)

URL: <https://dvc.org/>

This version control system is designed for managing machine learning models, data sets, and experiments efficiently.

Apache Airflow

URL: <https://airflow.apache.org/>

While primarily used for data orchestration and workflow automation, Airflow can also be leveraged for MLOps to schedule and monitor model training and deployment tasks.

Metaflow

URL: <https://metaflow.org/>

Developed by Netflix, Metaflow is a data science framework that assists in building and managing real-life data science projects. It allows easy integration of code, data, and models.

Knative

URL: <https://knative.dev/>

Not specific to MLOps, Knative is an open source platform that extends Kubernetes to support serverless workloads, making it useful for

managing serverless machine learning deployments.

CML (Continuous Machine Learning)

URL: <https://cml.dev/>

An open source library that integrates with Git and GitHub Actions to enable continuous integration and delivery for machine learning projects.

Seldon Core

URL: <https://www.seldon.io/tech/products/core/>

Built on Kubernetes, Seldon Core helps deploy and manage machine learning models at scale, allowing you to create production-ready inference graphs.

Apache Spark

URL: <https://spark.apache.org/>

While primarily a Big Data processing engine, Spark's MLlib library provides various tools for building and managing machine learning pipelines.

Research in the domain of MLOps can delve into identifying and analysing the best practices, methodologies and frameworks for integrating machine learning and DevOps effectively. This could involve studying existing MLOps tools, platforms, and practices to understand their strengths, weaknesses, and areas of improvement. Research may also explore techniques for automating model training, testing, and deployment pipelines.

Research could also address the security and governance challenges associated with deploying ML models in production environments. This may involve exploring methods to secure models, data, and infrastructure, as well as adhering to regulatory compliance.

Researchers can contribute to enhancing the efficiency, reliability, and ethical implications of deploying ML models in real world settings, making them more accessible and beneficial to society. **END** 

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